

Dr. Alex I. Malz
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Professional Appointments

2025– Barry M. Lasker Data Science Postdoctoral Fellow, Space Telescope Science Institute
2022–25 Project Scientist, Vera C. Rubin Observatory Legacy Survey of Space and Time (LSST)
Interdisciplinary Network for Collaborative Computing (LINCC) Frameworks
2025 Special Faculty, Dept. of Statistics & Data Science, Carnegie Mellon University (CMU)
2019–22 Postdoctoral Research Fellow, German Centre for Cosmological Lensing (GCCL)

Education

PhD 2020, Physics, New York University (NYU) Grad. School of Arts and Sciences (GSAS)
Thesis: Probabilistic analysis methods for cosmology using uncertainty-dominated photometric data
Advisor: Prof. David Hogg DOI: [dx.doi.org/10.5281/zenodo.3973536](https://doi.org/10.5281/zenodo.3973536)
MS 2014, Astronomy & Astrophysics, Pennsylvania State University (PSU)
BS 2011, Physics & History, California Institute of Technology (Caltech)

Awards

Research Corporation for Scientific Advancement Fellow 2025
Dance Your Ph.D. Contest Finalist 2019
NYU GSAS Ted Kousseff Fellow 2018–19; NYU GSAS MacCracken Fellow 2014–18
Department of Energy Office of Science Graduate Student Research SCGSR Fellow 2017
PSU Astronomy & Astrophysics Braddock/Roberts Fellow 2012–13
Caltech President’s Scholar 2006–11

Leadership & Scientific Collaborations

Rubin LSST Science Collaborations (SCs): *Uniquely, A.I.M. is active in all eight SCs.*
Informatics & Statistics SC (ISSC) Co-chair 2025–; Publication Committee 2024–;
ISSC Executive Council 2023–25; Charter Committee 2023; Member 2019–
Strong Lensing ; Solar System ; Stars, Milky Way & Local Volume SC Member 2024–
Galaxies SC; Transients & Variable Stars SC; Active Galactic Nuclei SC Member 2022–
Dark Energy SC (DESC) Builder 2020–, Full Member 2016–, Member 2016–;
DESC Photometric Redshifts Working Group Co-Convener 2019–21;
DESC Redshift Assessment Infrastructure Layers (RAIL) Topical Team Co-Lead 2021–24;
DESC Publication Bd. 2023–25; Collaboration Council 2018–20; Membership Comm. 2017–19
Nancy Grace Roman Space Telescope High Latitude Imaging Survey (HLIS) Project
Infrastructure Team (PIT) 2023–
Cosmological Adv. Surv. Telescope for Optical and ultraviolet Research (CASTOR) 2022–
Cosmostatistics Initiative (COIN) Member 2018–

Grants

Co-I, LSST Discovery Alliance (LSST-DA) EDI Grant 2024 ~ \$10⁴, DESC Sprint School
PI, DESC workshop 2023 ~ \$10⁴, RAIL Topical Team Sprint Week
PI, LSST-DA grant 2023 ~ \$10³, mentee travel to the Rubin Proj. and Community Wkshp.
PI, CMU McWilliams/PSC Seed Grant 2022–3 ~ \$10⁴, “Impossible Problems” seminars
PI, LSST Corporation (LSSTC) Enabling Sci. 2022 ~ \$10³, ISSC Ambassador student travel
PI, LSSTC Enabling Science 2021 ~ \$10⁴, “Stress-testing multimodal photo-*z* posteriors”
PI, DESC workshop 2019 ~ \$10⁴, Cluster Lensing Mass Modeling Sprint Week
Co-I, DESC workshop 2018 ~ \$10⁴, Cluster Lensing Mass Modeling Sprint Week
Co-I, NSF AST-1517237 2015 ~ \$1.5 × 10⁷, “New Probabilistic Methods for Obs. Cosmology”

Conferences, Workshops, Hackathons & Service

37 invited talks and 24 contributed presentations at conferences/workshops since 2016

3-time Astro Hack Week participant, 2-time COIN Residential Program participant

4-time SOC and 2-time LOC member; grant reviewer for NSF AAG and LINCC Incubator Prog.

Invited Talks (partial list)

2025

LSST Latin America Meeting, Keynote Talk

University of San Diego, Phys. & Biophys. Dept. Colloquium

2024

Ruhr University Bochum (RUB), Astronomy Colloquium

University of Washington, Astronomy Department Colloquium

Northwestern U., Cent. for Interdisc. Expl. and Res. in Astrophys. (CIERA) Theory

Mass. Inst. of Tech., Inst. for Artificial Intel. and Fund. Interactions (IAIFI)

2023

University of Delaware, Physics Seminar

University of Delaware, Electrical and Computer Engineering Colloquium

CMU, 3rd Nobel Turing Challenge Initiative Workshop

Northeastern University, Physics Colloquium

CMU, Statistical Methods for Physical Sciences (STAMPS) Colloquium

2022

CMU, Impossible Problems Interdisciplinary Wine-and-Cheese Seminar

Pan-Survey SED Forum, Photometric Redshifts

Lawrence Livermore National Laboratory, Lab-wide Seminar

2021

Ludwig-Maximilians-Universität Munich, Origins Cluster Data Science Lab. Seminar

American Astron. Soc. (AAS) #238 Meeting-in-a-Meeting, Machine Learning in Astro.

Princeton University, Institute for Advanced Study Cosmology Lunch Talk

Machine Learning Club, Debate Panel: Will ML Solve Photometric Redshifts?

2019

Berkeley Cen. for Cosm. Phys., Accurate Lensing in the Era of Precision Cosmo. Wkshp.

Royal Observatory Edinburgh, Institute for Astronomy Seminar

University of Washington, Data Intensive Res. in Ast. and Cosmology (DIRAC) Seminar

2018

Lawrence Livermore National Laboratory, Cosmology Seminar

Lawrence Berkeley Natl. Lab., Berkeley Cent. for Cosmological Phys. (BCCP) Seminar

Leiden University Lorentz Center, Colours of the Universe Workshop

CMU, ML + Time Domain Workshop

Laboratoire de Physique Corpusculaire de Clermont, Seminar

Laboratoire de Physique Subatomique et de Cosmologie de Grenoble, Colloquium

2017

Stony Brook University, Astronomy Colloquium

University of Toronto, Canadian Institute for Theoretical Astrophysics Seminar

Stanford University, Kavli Institute for Particle Astrophysics and Cosmology Tea

Tohoku University, Photo-z Workshop for Large Surveys

2016

University College London, Astronomy Seminar

Imperial College London, Astronomy Seminar

University of Pittsburgh, Photometric Redshifts for LSST Workshop

Research Supervision & Mentoring (★ indicates mentee-led publication)

2025– Grant Zhou★NeurIPS (2025); CMU Statistics & Data Science, PhD 2029

2024– Rachel Ouyang; CMU Physics→UChicago Physics, BS 2027

2024 Jenny Dong & Andrew Yuan; CMU Statistics & Data Science, BS 2027

2023–25 Alice Crafford★in DESC int.rev.; CMU Physics, BS 2025; U. Paris-Saclay, MS as Fullbright Scholar 2027

2021–22 Natalia Stylianou★PASP 134 1034 (2022); U. of Leicester, BS & MS 2022; Oxford U., PhD 2025

2021–22 Nicola Hunfeld; RUB Astronomy, MS 2022

2018 Zora Tung, Ted Singer, Dave Perrett & David Mykytyn; non-academic professionals

2018 Marin Hyatt & Lia Lubit; Hunter College High School 2020

Teaching**Instructor of Record**

CMU Stats & Data Sci. 219 Probability Th. & Rand. Processes Sp2025, 3 hr/wk, 50 students

PSU ASTRO 011 Elementary Astronomy Laboratory Sp2013, 3 hr/wk, 25 students

Invited Teacher

Bldg. a Phys. Understanding of Gal. Evol. w/Data-driven Astro. KITP-CCA, Wint2023, 3 hr

LSSTC Data Science Fellowship Program Northwestern U., Fall 2022, 12 hr

From Quarks to Cosmos with AI CMU, Summer 2021, 3 hr

LSST-DESC Dark Energy School U. Arizona, Winter 2020, 1.5 hr

Teaching Assistant

NYU PHYS-UA 7 The Universe: Its Nature and History Spring 2018

NYU PHYS-UA 15 Introduction to Cosmology Fall 2017

PSU ASTRO 120 The Big Bang Universe Spring 2014

PSU ASTRO 292 Astronomy of the Distant Universe Springs 2013, -14

PSU ASTRO 291 Astronomical Methods and the Solar System Falls 2012, -13

PSU ASTRO 001 Astronomical Universe Falls 2012, -13

Guest Lecturer

PSU ASTRO 485 Introduction to High-Energy Astronomy Fall 2013

PSU ASTRO 291 Astronomical Methods and the Solar System Fall 2012

PSU ASTRO 001 Astronomical Universe Fall 2012

Public Outreach

CMU Outreach Committee member 2023–25

STEM@CMU Panelist, Chief Science Officer Leadership training program Fall 2023

Judge, New York City Science and Engineering Fair finals event Springs 2018, -21, -22

Juror, German Young Physicists' Tournament North Rhine-Westphalia Winter 2021

Contestant, Dance Your Ph.D. tinyurl.com/aimalz-dance-your-phd Winter 2018–19

Produced, directed, choreographed, and danced an entertaining explanation of Malz & Hogg (2021)

Math Circle Speaker at Bridge to Enter Advanced Mathematics beammath.org Summer 2018

Designed and taught applied geometry lessons in the context of astronomy

Judge, New York City Science and Engineering Fair finals event Spring 2018, -21

Guest Speaker, Hunter College High School Science Club Spring 2016

Judge, Pennsylvania Junior Academy of Science finals event Spring 2013

Outreach Developer & Facilitator, PSU AstroFest Summers 2012, -13, -14

Created and implemented tie-dye-based activities teaching astronomy concepts tinyurl.com/astro-tie-dye

Publications (★ indicates (co-)lead authorship roles)

AIM has been awarded Builder Status within the LSST-DESC, which grants authorship rights on all DESC publications enabled by his work on photometric redshifts, time-domain classification, and service to the collaboration; however, he only co-authors papers to which he made *a direct scientific or technical contribution*.

43. A. Crafford, ★**A.I. Malz**, T. Zhang, R. Mandelbaum, O. Lynn, F. Berlfein, J. Cohen-Tanugi, J.F. Crenshaw, Q. Hang, I. Moskowitz, D. Oldag, S.J. Schmidt, Z. Yan, LSST-DESC. *sub. to MNRAS 15 January 2026*. “Diagnosing the Effects of Spectroscopic Training Set Imperfection on Photometric Redshift Performance” ([arXiv:2601.10797](#))
Lead author: conceptualization, funding acquisition, methodology, project administration, resources, supervision, validation, writing

42. The RAIL Team, J.L. van den Busch, E. Charles, J. Cohen-Tanugi, A. Crafford, J.F. Crenshaw, S. Dagoret, J. De Santiago, J. De Vicente, Q. Hang, B. Joachimi, S. Joudaki, J.B. Kalmbach, S. Liang, O. Lynn, ★**A.I. Malz**, R. Mandelbaum, G. Merz, I. Moskowitz, D. Oldag, J. Ruiz-Zapatero, M. Rahman, S.J. Schmidt, J. Scora, R. Shirley, B. Stölnzer, L. Toribio San Cipriano, L. Tortorelli, Z. Yan, T. Zhang, LSST-DESC. *accepted to OJA 30 December 2025*. “Redshift Assessment Infrastructure Layers (RAIL): Rubin-era photometric redshift stress-testing and at-scale production” ([arXiv:2505.02928](#))
Lead author: conceptualization, formal analysis, funding acquisition, investigation, methodology, project administration, software, supervision, validation, visualization, writing

41. L. Palaversa, E. Donev, Ž. Ivezić, K. Mrakovčić, N. Caplar, M. Jurić, T. Jurkić, S. Campos, M. DeLucchi, D. Jones, K. Malanchev, **A.I. Malz**, S. McGuire, B. Abel, L. Girardi, G. Pastorelli, M. Trabucchi, S. Zaggia, E. Acosta, C.L. Adair, J. Andrew, É. Aubourg, A.E. Bauer, W. Beebe, E.C. Bellm, R.D. Blum, M.T. Booth, A. Boucaud, D. Branton, D.L. Burke, D. Calabrese, J.L. Carlin, H-F. Chiang, Y. Choi, A.J. Connolly, S. Dagoret-Campagne, P.N. Daly, F. Daruich, G. Daubard, E. Dennihy, H. Drass, O. Eiger, A.M. Eisner, L.P. Guy, J. Hoblitt, P. Ingraham, F. Jammes, B.T. Jannuzi, M.J. Jee, T. Jenness, R.L. Jones, C. Juramy-Gilles, S.M. Kahn, Y. Kang, A. Kannawadi, L.S. Kelvin, I.V. Kotov, G. Kovacs, N.R. Kurita, T. Lange, D. Laporte, J.C. Lazarte, S. Liang, M. Lopez, N.B. Lust, M. Lutfi, O. Lynn, G. Mainetti, F. Menanteau, M. Miller, M. Moniez, N. Sedaghat, E. Nourbakhsh, H.Y. Park, J.R. Peterson, R. Plante, A. Plazas Malagón, M.N. Porter, K.A. Reil, V.J. Riot, A. Roodman, E.S. Rykoff, R.H. Schindler, J. Sebag, R.A. Shaw, A. Shugart, K.B. Siruno, J.A. Smith, J.D. Swinbank, J.G. Thayer, S. Thomas, R. Tighe, D.L. Tucker, M. Turri, E.K. Urbach, B. Van Klaveren, W. van Reeve, C.Z. Waters, B. Willman. *sub. to A&A 12 September 2025*. “photoD with Rubin’s Data Preview 1: first stellar photometric distances and deficit of faint blue stars. Stellar distances with Rubin’s DP1” ([arXiv:2512.24109](#))
Contributor: conceptualization

40. Z. Zhou, ★**A.I. Malz**, C. Schafer. 2025. Machine Learning and the Physical Sciences Workshop at the 39th Neural Information Processing Systems (NeurIPS) conference. “Improving and Assessing Astronomical Light Curve Classifiers with Classification Histories”. (NeurIPS)
Lead author: conceptualization, data curation, investigation, methodology, project administration, supervision, validation, visualization, writing

39. J.F. Crenshaw, B. Leistedt, M.L. Graham, C. Payerne, A.J. Connolly, E. Gawiser, T. Karim, **A.I. Malz**, J.A. Newman, M. Ricci, LSST-DESC. 2025. ApJS 281 2 54.
“Quantifying the Impact of LSST -band Survey Strategy on Photometric Redshift Estimation and the Detection of Lyman-break Galaxies”
Contributor: conceptualization, software

38. V.G. Shah, A. Gagliano, K. Malanchev, G.Narayan, **A.I. Malz**, LSST-DESC. 2025. ApJ 995 1 4. “ORACLE: A Real-Time, Hierarchical, Deep-Learning Photometric Classifier for the LSST”
Contributor: data curation, software
37. T. Zhang, E. Charles, J.F. Crenshaw, S.J. Schmidt, P. Adari, J. Gschwend, S. Mau, B. Andrews, E. Aubourg, Y. Bains, K. Bechtol, A. Boucaud, D. Boutigny, P. Burchat, J. Chevalier, J. Chiang, H.-F. Chiang, D. Clowe, J. Cohen-Tanugi, C. Combet, A. Connolly, S. Dagoret-Campagne, P.N. Daly, F. Daruich, G. Daubard, J. De Vicente, H. Drass, K. Fanning, E. Gawiser, M. Graham, L.P. Guy, Q. Hang, P. Ingraham, O. Ilbert, M. Jarvis, M.J. Jee, T. Jenness, A. Johnson, C. Juramy-Gilles, S.M. Kahn, J.B. Kalmbach, Y. Kang, A. Kannawadi, L.S. Kelvin, S. Liang, O. Lynn, N.B. Lust, M. Lutfi, **A.I. Malz**, R. Mandelbaum, S. Marshall, J. Meyers, M. Migliore, M. Moniez, J. Neveu, J.A. Newman, E. Nourbakhsh, D. Oldag, H. Park, S. Pelesky, A.A. Plazas Malagón, B. Quint, M. Rahman, A. Rasmussen, K. Reil, W. Roby, A. Roodman, C. Roucelle, M. Salvato, B. Sánchez, D. Sanmartim, R.H. Schindler, J. Scora, J. Sebag, N. Sedaghat, I. Sevilla-Noarbe, R. Shirley, A. Shugart, R. Solomon, D. Taranu, G. Thayer, L. Toribio San Cipriano, E. Urbach, Y. Utsumi, W. van Reeve, A. von der Linden, C.W. Walter, W.M. Wood-Vasey, J. Zuntz, LSST-DESC. *submitted to MNRAS 8 October 2025*. “Photometric Redshift Estimation for Rubin Observatory Data Preview 1 with Redshift Assessment Infrastructure Layers (RAIL)” (arXiv:2510.07370)
Contributor: software
36. Y.-H. Zhang, J. Zuntz, I. Moskowitz, E. Gawiser, K. Kuijken, M. Asgari, H. Hoekstra, **A.I. Malz**, Z. Yan, T. Zhang, LSST-DESC. *submitted to MNRAS 28 August 2025*. “Improved photometric redshift estimations through self-organising map-based data augmentation” (arXiv:2508.20903)
Contributor: funding acquisition, software
35. K. Malanchev, M. DeLucchi, N. Caplar, **A.I. Malz**, A. Alexov, E. Aubourg, A.E. Bauer, W. Beebe, E.C. Bellm, R.D. Blum, D. Branton, S. Campos, D. Calabrese, J.L. Carlin, Y. Choi, A. Connolly, M. Dai, P.N. Daly, F. Daruich, G. Daubard, F. Delgado, H. Drass, G. Fonseca Alvarez, E. Gangler, L.P. Guy, Ž Ivezić, F. Jammes, B.T. Jannuzi, T. Jenness, D. Jimenez, D.T. Jones, M. Jurić, S.M. Kahn, Y. Kang, A. Kannawadi, J. Kubica, L. Le Guillou, O. Lynn, R. Mandelbaum, M. Moniez, C.B. Morrison, S. McGuire, I.E. Ordenes, M.T. Patterson, A.A. Plazas Malagón, Y. Qiu, B. Quint, V. Radeka, W. van Reeve, V.J. Riot, B.O. Sánchez, D. Sanmartim, J. Sebag, A. Shugart, I.S. Sullivan, D.S. Taranu, E.K. Urbach, W.M. Wood-Vasey, T. Zhang. *submitted to OJA 1 July 2025*. “Variability-finding in Rubin Data Preview 1 with LSDB” (arXiv:2506.23955)
Contributor: validation, writing
34. B. Scott, ★**A.I. Malz**, R. Sorba. 2025. ApJ 985 2 227. “A holistic exploration of the potentially recoverable redshift information of Stage IV photometric galaxy surveys”
Lead author: conceptualization, formal analysis, investigation, methodology, project administration, software, supervision, validation, visualization, writing
33. ★**A.I. Malz**, M. Dai, K.A. Ponder, E.E.O. Ishida, S. Gonzalez-Gaitain, R. Durgesh, A. Krone-Martins, R.S. de Souza, N. Kennamer, S. Sreejith, L. Galbany. 2025. A&A 694 A130. “Are classification metrics good proxies for SN Ia cosmological constraining power?”
Lead author: conceptualization, formal analysis, investigation, methodology, software, validation, visualization, writing
32. I. Mendoza, A. Torchylo, T. Sainrat, A. Guinot, A. Boucaud, M. Paillasa, C. Avestruz, P. Adari, E. Aubourg, B. Biswas, J. Buchanan, P. Burchat, C. Doux, R. Joseph, S. Kamath,

- A.I. Malz**, G. Merz, H. Miyatake, C. Roucelle, T. Zhang, LSST-DESC. 2025. OJA 8 “The Blending ToolKit: A simulation framework for evaluation of galaxy detection and deblending”
Contributor: validation, visualization
31. L. Palaversa, Z. Ivezić, N. Caplar, K. Mrakovic, B. Abel, O. Razim, F. Matkovic, C. Yablonski, T. Saric, T. Juric, S. Campos, M. DeLucchi, D. Jones, K. Malanchev, **A.I. Malz**, S. McGuire, M. Juric. 2025. AJ 169 119. “PhotoD with LSST: Stellar Photometric Distances Out to the Edge of the Galaxy”
Contributor: conceptualization, software
30. G. Merz, X. Liu, S. Schmidt, **A.I. Malz**, T. Zhang, D. Branton, C.J. Burke, M. Delucchi, Y.S. Ejjagiri, J. Kubica, Y. Liu, O. Lynn, D. Oldag, LSST-DESC. 2025. OJA 8 40. “DeepDISC-photoz: Deep Learning-Based Photometric Redshift Estimation for Rubin LSST”
Contributor: conceptualization, software
29. J. Lee, M. Sako, R. Kessler, **A.I. Malz**. 2024. ApJ 977 2 199. “Astrometric Redshifts of Supernovae”
Contributor: data curation, software, writing
28. Q. Hang, B. Joachimi, E. Charles, J.F. Crenshaw, P. Larsen, **A.I. Malz**, S. Schmidt, Z. Yan, T. Zhang, LSST-DESC. 2024. MNRAS 535 4. “Impact of survey spatial variability on galaxy redshift distributions and the cosmological 3×2 -point statistics for the Rubin Legacy Survey of Space and Time (LSST)”
Contributor: conceptualization, funding acquisition, software, writing
27. ★**A.I. Malz**, F. Lanusse, J.F. Crenshaw, B. Scott, M.L. Graham, X. Li. *accepted to ApJS 17 September 2024*. “An information-based metric for observing strategy optimization, demonstrated in the context of photometric redshifts with applications to cosmology”
Lead author: conceptualization, data curation, formal analysis, funding acquisition, investigation, methodology, software, validation, visualization, writing
26. K.M. de Soto, A. Villar, E. Berger, S. Gomez, G. Hosseinzadeh, D. Branton, S. Campos, M. DeLucchi, J. Kubica, O. Lynn, K. Malanchev, **A.I. Malz**. 2024. ApJ 974 2 169. “Superphot+: Realtime Fitting and Classification of Supernova Light Curves”
Contributor: conceptualization, methodology
25. J.F. Crenshaw, J.B. Kalmbach, A. Gagliano, Z. Yan, A.J. Connolly, **A.I. Malz**, S.J. Schmidt. 2024. AJ 168 80 “Probabilistic Forward Modeling of Galaxy Catalogs with Normalizing Flows”
Contributor: conceptualization, funding acquisition, validation
24. D. Oldag, M. DeLucchi, W. Beebe, D. Branton, S. Campos, C.O. Chandler, C. Christofferson, A. Connolly, J. Kubica, O. Lynn, K. Malanchev, **A.I. Malz**, R. Mandelbaum, S. McGuire, C. Wenneman. 2024. RNAAS 8 5 141. “A Python Project Template for Healthy Scientific Software”
Contributor: conceptualization, software, writing
23. I. Moskowitz, E. Gawiser, J.F. Crenshaw, B.H. Andrews, **A.I. Malz**, S. Schmidt. 2024. ApJL 967 L6. “Improving Photometric Redshift Estimates with Training Sample Augmentation”
Contributor: software

22. A. Gagliano, G. Contardo, D. Foreman-Mackey, **A.I. Malz**, P.D. Aleo. 2023. ApJ 954 6. “First Impressions: Early-Time Classification of Supernovae using Host Galaxy Information and Shallow Learning”
Contributor: data curation, resources, software, writing
21. R. Hložek, ★**A.I. Malz**, K.A. Ponder, M. Dai, G. Narayan, E.E.O. Ishida, T. Allam Jr., A. Bahmanyar, R. Biswas, L. Galbany, S.W. Jha, D.O. Jones, R. Kessler, M. Lochner, A.A. Mahabal, K.S. Mandel, J.R. Martinez-Galarza, J.D. McEwen, D. Muthukrishna, H.V. Peiris, C.M. Peters, C.N. Setzer. 2023. ApJS 267 25. “Results of the Photometric LSST Astronomical Time-series Classification Challenge (PLAsTiCC)”
Lead author: conceptualization, formal analysis, investigation, methodology, visualization, writing
20. M. Lokken, A. Gagliano, G. Narayan, R. Hložek, R. Kessler, J.F. Crenshaw, L. Salo, C.S. Alves, D. Chatterjee, M. Vincenzi, **A.I. Malz**. 2023. MNRAS 520 2. “The Simulated Catalogue of Optical Transients and Correlated Hosts (SCOTCH)”
Contributor: conceptualization, data curation, methodology
Please pardon the apparent paucity of publications in the preceding two years as a result of the global pandemic limiting collaboration opportunities.
19. N. Stylianou, ★**A.I. Malz**, P. Hatfield, J.F. Crenshaw, J. Gschwend. 2022. PASP 134 1034. “The sensitivity of GPz estimates of photo-z posterior PDFs to realistically complex training set imperfections”
Lead author: conceptualization, data curation, formal analysis, funding acquisition, investigation, methodology, software, supervision, validation, visualization, writing
18. ★**A.I. Malz**, D.W. Hogg. 2022. ApJ 928 127. “How to obtain the redshift distribution from probabilistic redshift estimates”
Lead author: conceptualization, data curation, formal analysis, funding acquisition, investigation, methodology, software, validation, visualization, writing
17. M. Agüena, C. Avestruz, C. Combet, S. Fu, R. Herbonnet, ★**A. I. Malz**, M. Penna-Lima, M. Ricci, S. D. P. Vitenti, L. Baumont, H. Fan, M. Fong, M. Ho, M. Kirby, C. Payerne, D. Boutigny, B. Lee, B. Liu, T. McClintock, H. Miyatake, C. Sifón, A. von der Linden, H. Wu, M. Yoon. 2021. MNRAS 508 4 6092. “CLMM: a LSST-DESC Cluster weak Lensing Mass Modeling library for cosmology”
Lead author: conceptualization, funding acquisition, methodology, project administration, software, validation, writing
16. B. Dey, J.A. Newman, B.H. Andrews, R. Izbicki, A.B. Lee, D. Zhao, M.M. Rau, **A.I. Malz**. 2021. Fourth Workshop on Machine Learning and the Physical Sciences at the 35th Neural Information Processing Systems (NeurIPS) conference. “Re-calibrating Photometric Redshift Probability Distributions Using Feature-space Regression” ([arXiv:2110.15209](https://arxiv.org/abs/2110.15209))
Contributor: conceptualization, methodology, supervision, writing
15. J. Zuntz, F. Lanusse, ★**A.I. Malz**, A.H. Wright, A. Slosar, B. Abolfathi, D. Alonso, A. Bault, C.R. Bom, M. Brescia, A. Broussard, J.-E. Campagne, S. Cavaoti, E.S. Cypriano, B.M.O. Fraga, E. Gawiser, E.J. Gonzalez, D. Green, P. Hatfield, K. Iyer, D. Kirkby, A. Nicola, E. Nourbakhsh, A. Park, G. Teixeira, K. Heitmann, E. Kovacs, Y.-Y. Mao. 2021. OJA 4. “The LSST-DESC 3x2pt Tomography Optimization Challenge”
Lead author: conceptualization, formal analysis, methodology, visualization, writing
14. ★**A.I. Malz**. 2020. PRD 103 083502. “How NOT to obtain the redshift distribution from probabilistic redshift estimates”
Sole author: conceptualization, formal analysis, investigation, methodology, project administration, resources, writing

13. B. Moews, M.S. Schmitz, A.J. Lawler, J. Zuntz, **A.I. Malz**, R.S. de Souza, R. Vilalta, A. Krone-Martins, E.E.O. Ishida. 2020. MNRAS 500 1 859. “[Ridges in the Dark Energy Survey for cosmic trough identification](#)”
Contributor: conceptualization, methodology, writing
12. S.J. Schmidt, ★**A.I. Malz**, J.Y.H. Soo, I.A. Almosallam, M. Brescia, S. Caviuoti, J. Johen-Tanugi, A.J. Connolly, J. DeRose, P.E. Freeman, M.L. Graham, K.G. Iyer, M.J. Jarvis, J.B. Kalmbach, E. Kovacs, A.B. Lee, G. Longo, C.B. Morrison, J.A. Newman, E. Nourbakhsh, E. Nuss, T. Pospisil, H. Tranin, R.H. Wechsler, R. Zhou, R. Izbicki. 2020. MNRAS 499 2 1587. “[Evaluation of probabilistic photometric redshift estimation approaches for LSST](#)”
Lead author: conceptualization, formal analysis, investigation, methodology, project administration, software, supervision, validation, visualization, writing
11. N. Kenmamer, E.E.O. Ishida, S. Gonzalez-Gaitan, R.S. de Souza, A. Ihler, K. Ponder, R. Vilalta, A. Moller, D.O. Jones, M. Dai, A. Krone-Martins, B. Quint, S. Sreejith, **A.I. Malz**, L. Galbany. 2020. IEEE Symposium Series on Computational Intelligence. “[Active learning with RESSPECT: Resource allocation for extragalactic astronomical transients](#)” ([arXiv:2010.05941](#))
Contributor: conceptualization, methodology
10. N. Dalmaso, T. Pospisil, A.B. Lee, R. Izbicki, P.E. Freeman, **A.I. Malz**. 2019. As. & Com. 20 100362. “[Conditional Density Estimation Tools in Python and R with Applications to Photometric Redshifts and Likelihood-Free Cosmological Inference](#)”
Contributor: data curation, writing
9. B. Moews, R.S. de Souza, E.E.O. Ishida, **A.I. Malz**, C. Heneka, R. Vilalta, J. Zuntz. 2019. PRD 99 123529. “[Stress testing the dark energy equation of state imprint on supernova data](#)”
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8. T. Cantat-Gaudin, A. Krone-Martins, N. Sedaghat, A. Farahi, R.S. de Souza, R. Skolidis, **A.I. Malz**, S. Macedo, B. Moews, C. Jordi, A. Moitinho, A. Castro-Ginard, E.E.O. Ishida, C. Heneka, A. Boucaud, A.M.M. Trindade. 2019. A&A 624 A126. “[Gaia DR2 unravels incompleteness of nearby cluster population: New open clusters in the direction of Perseus](#)”
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Lead author: conceptualization, data curation, formal analysis, investigation, methodology, project administration, software, supervision, validation, visualization, writing
6. C. Chang, M. Wang, S. Dodelson, T. Eifler, C. Heymans, M. Jarvis, M.J. Jee, S. Joudaki, E. Krause, **A.I. Malz**, R. Mandelbaum, I. Mohammed, M. Schneider, M. Simet, M.A. Troxel, J. Zuntz. 2018. MNRAS 482 3 3696. “[A Unified Analysis of Four Cosmic Shear Surveys](#)”
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1. A. Hagen, R. Ciardullo, C. Gronwall, V. Acquaviva, J. Bridge, G.R. Zeimann, G.A. Blanc, N.A. Bond, S.L. Finkelstein, M. Song, E. Gawiser, D.B. Fox, H. Gebhardt, **A.I. Malz**, D.P. Schneider, N. Drory, K. Gebhardt, G.J. Hill. 2014. ApJ 786 1 59. “Spectral Energy Distribution Fitting of HETDEX Pilot Survey Lyman-alpha Emitters in COSMOS and GOODS-N”
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Published Software

All of AIM’s code, including work in progress, is publicly available on his own GitHub or those of his collaborators and the collaborations in which he is active; these are the codes with formal releases accompanied by DOIs.

- LSST-DESC RAIL Topical Team (led by **A.I. Malz**). 2023. “RAIL”. DOI:10.5281/zenodo.7017551. <http://github.com/LSSTDESC/rail>
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Non-standard Publications

- I. Andreoni, et al. (incl. **A.I. Malz** 2024. *whitepaper*. “Rubin ToO 2024: Envisioning the Vera C. Rubin Observatory LSST Target of Opportunity program” ([arXiv:2411.04793](#))
Contributor: conceptualization, investigation, writing
- K. Breivik, et al. (incl. **A.I. Malz**) 2022. *whitepaper*. “From Data to Software to Science with the Rubin Observatory LSST” ([arXiv:2208.02781](#))
Contributor: conceptualization, investigation, writing
- LSST-DESC, et al. (incl. **A.I. Malz**) 2020. *LSST-DESC Research Note*. “The LSST-DESC Science Roadmap”
Contributor: conceptualization, project administration, writing
- T. Allam Jr., A. Bahmanyar, R. Biswas, M. Dai, L. Galbany, R. Hložek, E.E.O. Ishida, S.W. Jha, D.O. Jones, R. Kessler, M. Lochner, A.A. Mahabal, **A.I. Malz**, K.S. Mandel, J.R. Martinez-Galarza, J.D. McEwen, D. Muthukrishna, G. Narayan, H.V. Peiris, C.M. Peters, K. Ponder, C.N. Setzer. *LSST-DESC Research Note*. 2018. “The Photometric LSST Astronomical Time-series Classification Challenge (PLAsTiCC): Data set” ([arXiv:1810.00001](#))
Contributor: conceptualization, methodology, software
- A.I. Malz**, et al. 2018. “Dance Your Ph.D. 2018/9: Probabilistic methods for cosmological analysis with uncertainty-dominated data” (educational music video)
Lead author: conceptualization, funding acquisition, methodology (choreography), project administration (production), resources (costumes), software (video editing & web maintenance), supervision, visualization
- P.J. Marshall, et al. (incl. **A.I. Malz**) 2017. *whitepaper*. “Science-Driven Optimization of the LSST Observing Strategy” ([arXiv:1708.04058](#))
Contributor: conceptualization, methodology, writing
- A.I. Malz**. 2017. Cooper Square Review. “Going nowhere fast” (science communication essay)
Sole author: conceptualization, writing

Citeable Presentations

- M. DeLucchi, S. Campos, K. Malanchev, S. McGuire, N. Caplar, M. Juric, J. Kubica, M. West, D. Branton, O. Lynn, **A.I. Malz**, A. Connolly, R. Mandelbaum. 2025. American Astronomical Society, AAS Meeting #245, id. 471.02 “HATS and LSDB: LINCC Frameworks software for analysis of large catalogs” (contributed poster)
- K. Malanchev, W. Beebe, D. Branton, S. Campos, N. Caplar, A. Connolly, M. DeLucchi, M. Juric, J. Kubica, O. Lynn, **A.I. Malz**, R. Mandelbaum, S. McGuire, M. West. 2025. American Astronomical Society, AAS Meeting #245, id. 424.08. “Large-scale analysis of light-curve catalogs using LINCC Frameworks’ nested-pandas and LSDB” (contributed poster)
- M. Dai, M. DeLucchi, J. Kubica, O. Lynn, K. Malanchev, **A.I. Malz**, M. Tauraso. 2025. American Astronomical Society, AAS Meeting #245, id. 354.02. “TDAstro – A LINCC Frameworks Time-Domain Forward-Modeling Package for the Rubin Era” (contributed poster)
- S. McGuire, K. Malanchev, D. Branton, S. Campos, N. Caplar, A. Connolly, M. DeLucchi, M. Juric, J. Kubica, O. Lynn, **A.I. Malz**, R. Mandelbaum, M. West. 2025. American Astronomical Society, AAS Meeting #245, id. 259.03 “Analyzing the geometry of HEALPix tile boundaries to enable scalable processing of large scale catalogs” (contributed poster)
- K. Malanchev, W. Beebe, D. Branton, N. Caplar, A. Connolly, M. Dai, M. DeLucchi, J. Kubica, **A.I. Malz**, S. McGuire, D. Oldag. 2024. American Astronomical Society, AAS Meeting #243, id. 261.27 “LINCC Frameworks - Time series Analysis & Processing Engine (TAPE) for time-domain survey analysis” (contributed poster)
- M. West, J. Kubica, S. McGuire, S. Wyatt, M. DeLucchi, M. Juric, S. Stetzler, C. Slater, A. Connolly, R. Mandelbaum, **A.I. Malz**, G. Amvrosiadis, S. Campos, K. Malanchev. 2024. American Astronomical Society, AAS Meeting #243, id. 261.12 “LINCC Frameworks - HiPSCat Margin Caching” (contributed poster)
- S. Wyatt, M. Juric, S. McGuire, M. DeLucchi, M. West, K. Malanchev, C. Slater, S. Stetzler, A.

- Connolly, R. Mandelbaum, J. Kubica, **A.I. Malz**, S. Campos, G. Amvrosiadis. 2024. American Astronomical Society, AAS Meeting #243, id. 261.09 “LINCC - HiPSCat and LSDB: Joint Distributed Analysis of LSST-Scale Datasets” (contributed poster)
- A. Wasserman, J. Cohen-Tanugi, M. Dai, R. Hlozek, E.E.O. Ishida, H. Kelly, R. Kessler, A. Kim, R. Knop, K. Malanchev, **A.I. Malz**, G. Narayan, S. Thorp. 2024. American Astronomical Society, AAS Meeting #243, id. 232.02 “Uncovering Transient Physics and Optimizing Cosmological Inference with a Recommendation Engine for Rapid-Response Spectroscopy” (contributed poster)
- A.I. Malz** & the ELAsTiCC (ELAsTiCC) Team. 2023. American Astronomical Society, AAS Meeting #241, id. 117.04 “ELAsTiCC: Metrics of probabilistic classifications of the alert stream” (contributed talk)
- A.I. Malz** & the LSST-DESC RAIL Team. 2023. American Astronomical Society, AAS Meeting #241, Astronomy and Cloud Computing Special Session, id. 358.01. “All aboard! A LINCC Framework for extragalactic science using RAIL” (contributed poster)
- M. Juric, S. Stetzler, C. Slater, P. Zecevic, K. Boone, S. Wyatt, M. Delucchi, M. West, C. Christofferson, A. Connolly, R. Mandelbaum, J. Kubica, G. Amvrosiadis, **A.I. Malz**. 2023. American Astronomical Society, AAS Meeting #241, id. 312.03 “Reducing the optical depth: Solutions for an astronomer-friendly Cloud” (contributed talk)
- D. Bektsev, J. Kubica, M. West, C. Christofferson, B. Kalmbach, H. Smotherman, P. Bernardinelli, P. Whidden, D. Branton, M. Delucchi, D. Oldag, O. Lynn, **A.I. Malz**, C. Chandler, S. Wyatt, R. Mandelbaum, A. Connolly, M. Juric, G. Amvrosiadis, M. Wood-Vasey, J. Sokolski, C. Schafer, J. Newman. 2023. American Astronomical Society, AAS Meeting #241, id. 105.07 “LINCC Frameworks — KBMOD” (contributed poster)
- S. Wyatt, M. Juric, S. Stetzler, C. Slater, A. Connolly, M. DeLucchi, M. West, R. Mandelbaum, J. Kubica, G. Amvrosiadis, C. Christofferson, **A.I. Malz**. 2023. American Astronomical Society, AAS Meeting #241, id. 105.06 “LINCC — HIPSCat and LSD2: Joint Distributed Analysis of LSST-Scale Datasets” (contributed poster)
- N. Caplar, J. Kubica, M. West, D. Branton, M. Delucchi, C. Christofferson, D. Oldag, O. Lynn, **A.I. Malz**, C. Chandler, S. Wyatt, R. Mandelbaum, A. Connolly, M. Juric, G. Amvrosiadis, M. Wood-Vasey, J. Sokolski, C. Schafer, J. Newman. 2023. American Astronomical Society, AAS Meeting #241, id. 105.05 “LINCC Frameworks — Time series” (contributed poster)
- A. Connolly, R. Mandelbaum, J. Kubica, M. Juric, S. Wyatt, N. Caplar, C. Chandler, **A.I. Malz**, D. Branton, D. Oldag, M. West, C. Christofferson, J. Newman, M. Wood-Vasey, J. Sokoloski, G. Amvrosiadis, M. Delucchi, O. Lynn, C. Schafer. 2023. American Astronomical Society, AAS Meeting #241, id. 105.04 “The LINCC Frameworks Initiative” (invited talk)
- A.I. Malz**. 2021. American Astronomical Society, AAS Meeting #238, Machine Learning in Astronomy Meeting-in-a-Meeting, id. 103.02. “Proceed with caution: how, and how not, to use machine learning to probe cosmology” (invited talk)
- A.I. Malz**, F. Lanusse, J.F. Crenshaw, M.L. Graham. 2021. American Astronomical Society, AAS Meeting #238, id. 230.04. “**TheLastMetric**: an information-based observing strategy metric for photometric redshifts, cosmology, and more” (contributed poster)
- J.F. Crenshaw, J.B. Kalmbach, **A.I. Malz**, A.J. Connolly. 2021. American Astronomical Society, AAS Meeting #238, id. 230.01. “PZFlow: normalizing flows for cosmology, with applications to forward modeling galaxy photometry” (contributed poster)
- A.I. Malz**. 2021. American Astronomical Society, AAS Meeting #237, LSST-DESC Special Session, id. 443.05. “The DESC Photometric Redshifts Working Group: Challenges & Opportunities” (contributed talk)
- A.I. Malz**. 2019. American Astronomical Society, AAS Meeting #233, Surveys & Large Programs, id. 313.05D. “Probabilistic data analysis methods for large photometric surveys” (contributed dissertation talk)
- A.I. Malz**. 2019. American Astronomical Society, AAS Meeting #233, Larger Efforts in

- Education & Public Outreach, id. 212.05. “The Photometric LSST Astronomical Time Series Classification Challenge (PLAsTiCC): challenge design and evaluation criteria” (contributed talk)
- C.M. Peters, **A.I. Malz** & R. Hlozek. 2018. American Astronomical Society, AAS Meeting #231, id. 245.03. “Supernova Cosmology Inference with Probabilistic Photometric Redshifts” (contributed poster)
- A.I. Malz** & S. Shandera. 2014. American Astronomical Society, AAS Meeting #223, id. 456.04. “Probing Gravity in the High-Redshift Universe with HETDEX” (contributed poster)
- A.I. Malz**, R. Rich & S. Lepine. 2009. American Astronomical Society, AAS Meeting #213, id. 602.04. “Low-mass Binaries in the Galactic Halo Resolved by Adaptive Optics” (contributed poster)